

TITLE:

FTP Brute Force Detection and Prevention Using Wireshark, Hydra and iptables.

1. Introduction

In this project I simulated a brute force attack on an FTP server using Kali Linux as the attacker and Ubuntu as the target, The attack was detected using Wireshark and the malicious IP was blocked using firewall rules (iptables).

2. Tools & Setup

- Attacker: Kali Linux (Hydra)
- Target: Ubuntu (vsftpd FTP server)
- Monitoring Tool: Wireshark (on Ubuntu)
- Firewall: iptables (on Ubuntu)
- Network mode: bridge adapter

3. FTP Server Configuration on Ubuntu

-Installed vsftpd using:

```
sudo apt install vsftpd
```

-Enabled and started the service using:

```
Sudo systemctl enable vsftpd
```

```
Sudo systemctl start vsftpd
```

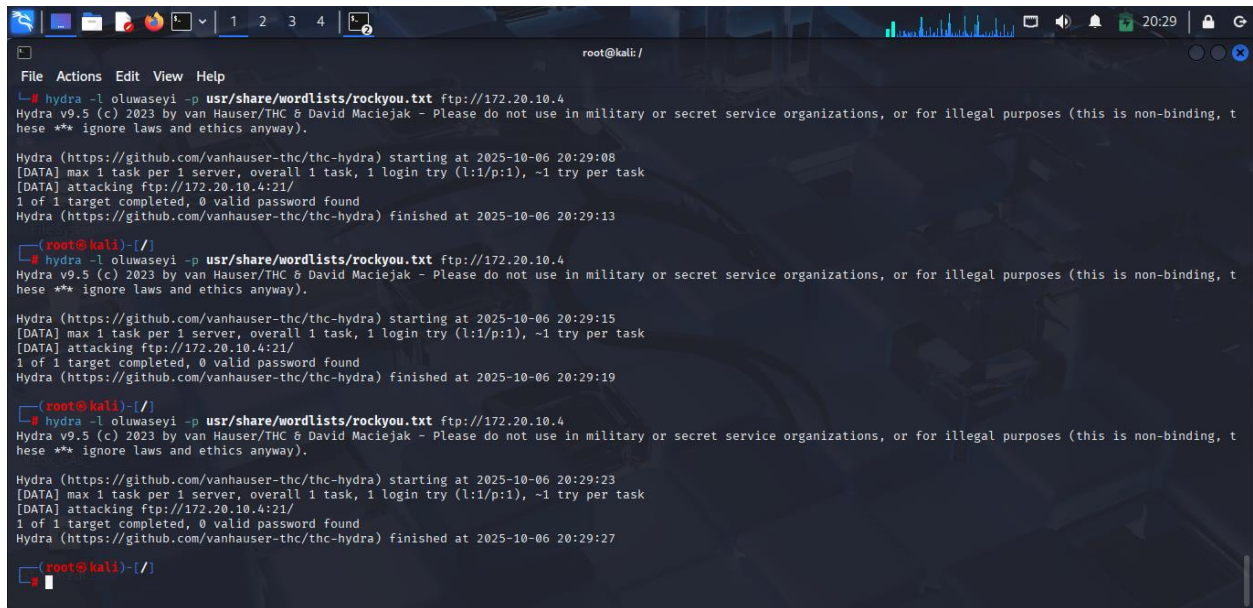
-Created an FTP user with a password using:

```
sudo adduser oluwaseyi
```

4. Attack Simulation with Hydra (Kali)

I simulated an attack using Hydra on kali to brute force the FTP login using the following command:

```
hydra -l oluwaseyi -p usr/share/wordlists/rockyou.txt  
ftp://172.20.10.4
```



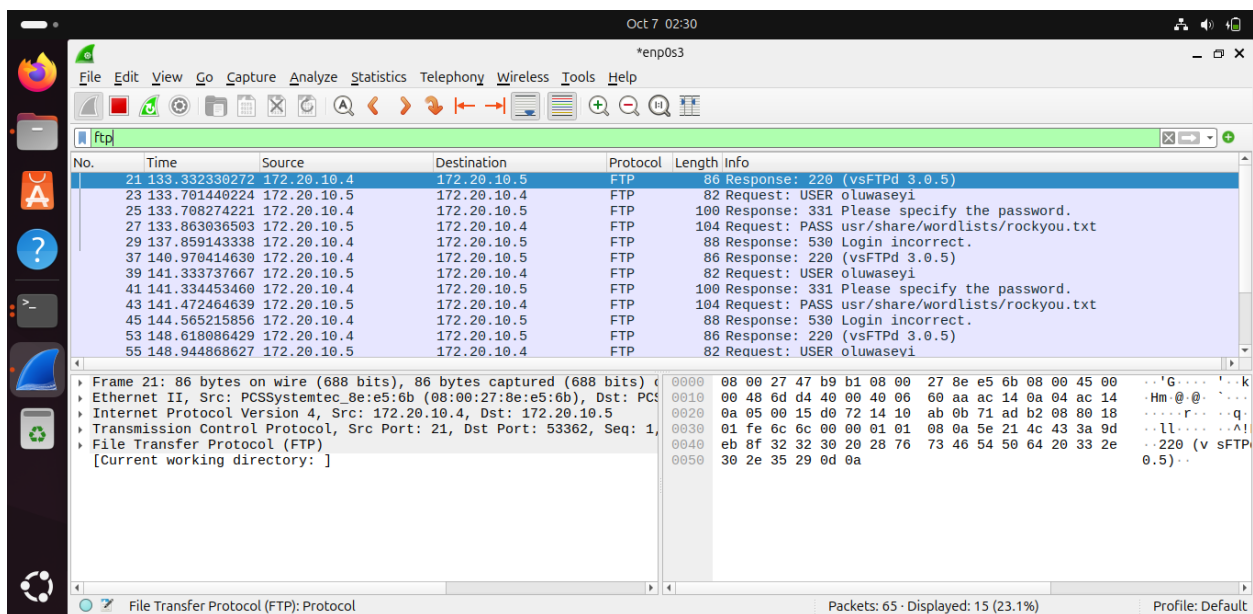
```
root@kali: /  
File Actions Edit View Help  
# hydra -l oluwaseyi -p usr/share/wordlists/rockyou.txt ftp://172.20.10.4  
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, t  
hese *** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-10-06 20:29:08  
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try per task  
[DATA] attacking ftp://172.20.10.4:21/  
1 of 1 target completed, 0 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-10-06 20:29:13  
  
# hydra -l oluwaseyi -p usr/share/wordlists/rockyou.txt ftp://172.20.10.4  
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, t  
hese *** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-10-06 20:29:15  
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try per task  
[DATA] attacking ftp://172.20.10.4:21/  
1 of 1 target completed, 0 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-10-06 20:29:19  
  
# hydra -l oluwaseyi -p usr/share/wordlists/rockyou.txt ftp://172.20.10.4  
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, t  
hese *** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-10-06 20:29:23  
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try per task  
[DATA] attacking ftp://172.20.10.4:21/  
1 of 1 target completed, 0 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-10-06 20:29:27  
  
#
```

5. Traffic Capture with Wireshark (Ubuntu)

On Ubuntu, Wireshark was used to monitor network interfaces I applied the following filter to view only ftp packets:

Used filter: ftp

I Observed multiple login attempts which is an evidence of brute force



6. ATTACK MITIGATION

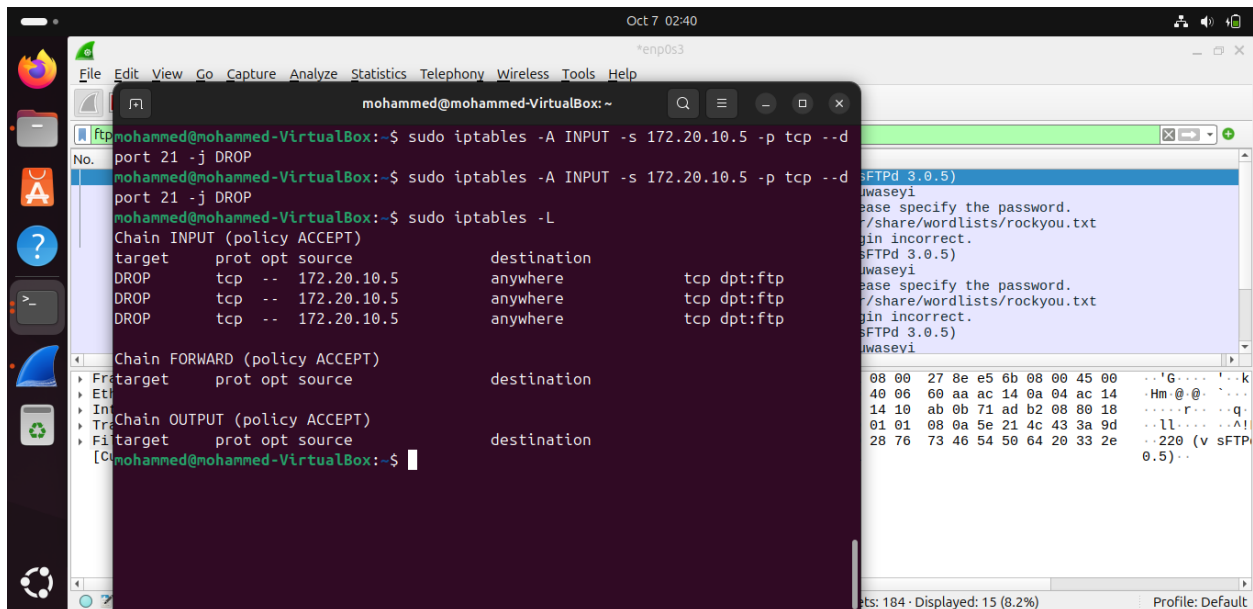
To mitigate the attack, we used the following iptables rule on the ubuntu machine:

```
sudo iptables -A INPUT -s 172.20.10.3 -p tcp --dport 21 -j DROP
```

This rule drops all incoming ftp packets from attacker's ip.

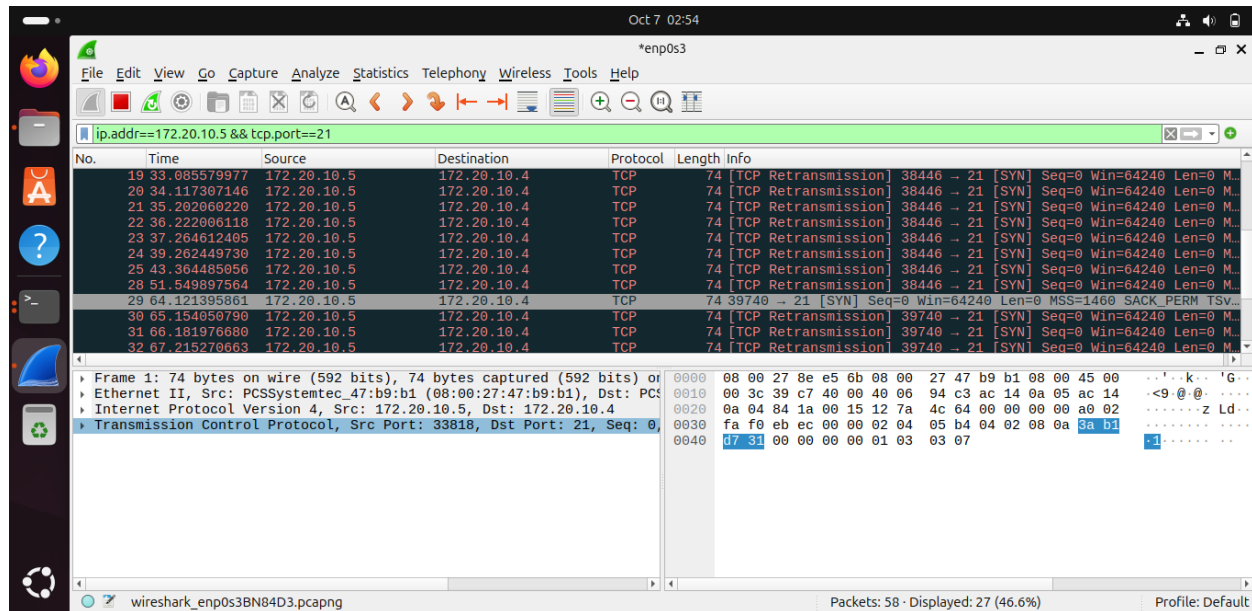
To verify the rule was applied I used the following command:

```
Sudo iptables -L
```



7. Verifying Block in Wireshark

I Re-ran the attack from Kali to Capture traffic again on Ubuntu, i Observed multiple TCP Retransmission packets showing that the firewalls successfully dropped the connection attempt



8. Conclusion

I Successfully simulated a brute force attack, Captured and analyzed it with Wireshark ,Mitigated the threat by blocking the attacker's IP with iptables and Verified the block through Wireshark packet analysis. This forms a key part of SOC analyst responsibility in real world environment.

Author: Mohammed Ndagi, Date: October 2025.

